

# SELF-EVOLUTION SHOULD NOT DEPEND ON HOW SKILLS ARE SPLIT: AN EXACT CERTIFICATE FOR SKILL-TAXONOMY REPRESENTATION INVARIANCE

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## ABSTRACT

Skill-using agents do not share one access schedule: some prefetch skills, some retrieve each turn, and others progressively load subskills or references. We ask a representation-invariance question at the common interface: if the underlying semantic candidate support is unchanged, should the semantic capability crossing a skill-access boundary change only because the same capability is packaged differently? We call this *Skill-Taxonomy Representation Invariance* (STRI). Runtime STRI compares the semantic projection of exposed capability under matched pre-access state, access request, mechanism, budget, and stochastic state while changing only the physical package realization. Separately, for package-only control we define  $R^*(A; q) = \min\{t : q \leq Aw \leq tq, w, t \geq 0\}$ ;  $R^* = 1$  iff the representation-independent target  $q$  lies in the package support cone, while  $R^* > 1$  certifies an irreducible structural residual for that control class. Released Skill-SP exhibits both residual and equalizable support geometries, and SkillRL shows identity-sensitive retrieval under finite budget. In one frozen AutoSkill P19 prefetch substrate, exact-semantic splitting changes the model-visible skill set and a mechanically defined downstream behavior, with ID-placebo, quotient, and mediator add-back controls isolating the path. Outcome-blind heldout qualification reproduces local retrieval sensitivity in 9/9 units, but a preregistered 2-unit/8-run behavior pilot provides no split-specific propagation evidence and stops. Thus the current evidence supports access-level representation sensitivity plus one bounded downstream witness; behavioral propagation beyond P19 is not established. The structural certificate and runtime evidence are parallel views of the same representation mismatch, not one causal algorithm or a general utility theorem.

## 1 INTRODUCTION

Skill-using agents increasingly externalize reusable procedure into persistent artifacts rather than foundation-model weights alone. Those artifacts need not be flat prompts: a skill may be one record, a directory bundle, or a root with references and subskills that are exposed progressively as execution unfolds (Bai et al., 2026a;b). Access schedules also differ: one system may prefetch skills before acting, another may retrieve every turn, and another may load a subskill only after the controller requests it. The common systems question is therefore not “what is the Top- $k$ ?” but what semantic capability actually crosses the skill-access boundary into the actor-visible or callable interface.

Physical packaging can nevertheless enter control. Skill-SP (Huang et al., 2026) samples a package identity before generating a training task, SkillRL (Xia et al., 2026) gives fresh skill IDs priority inside a fixed retrieval budget, and AutoSkill exposes a package-level search seam. Consider two representations with the same underlying semantic candidate support: one stores a capability once, another adds an exact-semantic identity; one groups unchanged primitives behind a macro package, another exposes them separately. If the access request, mechanism, resources, and semantic payload are fixed, should the actor receive a different semantic capability solely because package count, identity, or partition changed?

Prior work establishes that skill availability, presentation, retrieval, composition, compression, and exposure matter (Xu & Wu, 2026; Jiang et al., 2026; Gao, 2026; Bai et al., 2026a;b; Singh & Joachims, 2018; Biega et al., 2018); Sec. 2 separates our estimand. STRI asks whether the *access interface itself* is invariant after semantic content/support is fixed and only physical package realization changes. Its structural specialization then asks whether arbitrary package mass can realize a representation-independent semantic target.

**Three questions.** We separate three claims that are easy to conflate. **RQ1, Local STRI:** can semantics-preserving package realization change the semantic capability admitted at a frozen access boundary? **RQ2, structural boundary:** when package-level control cannot realize a semantic target, is the obstruction support geometry rather than duplicate count, overlap prevalence, or poor scalar tuning? **RQ3, propagation:** if local semantic access changes, does that difference propagate to executed behavior? The third claim is strictly stronger than the first two and is reported fail-closed.

**Released evidence and boundary.** Skill-SP Level-1 has 314 validator-covered tasks, 183 with multiple memberships; its threshold-0.33 name+description duplicate filter misses all five observed specific-generic support-overlap pairs, and exact-support min-cover pruning still leaves 71 overlapping tasks. In SkillRL, 11/12 exact fresh-ID clone targets change semantic retrieval over 223 ALFWorld descriptions and 5/12 reduce unique returned contents. These systems establish representation sensitivity, not its structural cause: a separate logical domain has 127/128 overlapping tasks yet neutral  $R^* = 1$ .

## Contributions.

1. **A skill-access-boundary invariant.** We define STRI at the interface where a fixed access mechanism exposes skill information or callable capability to an actor. Physical package identity, count, and partition may change; the semantic candidate support and semantic payload do not. The contribution is this representation-invariance audit object, not a new retriever, deduplicator, or diversity heuristic.
2. **An exact structural certificate.** For the package-only specialization, quotient factorization characterizes exact-refinement invariance and  $R^*(A; q)$  decides target realizability in the frozen support geometry. Semantic-first selection marks an action-basis escape; we reuse classical convex feasibility/duality rather than claim a new LP or cone theorem.
3. **Layered system evidence with a negative propagation boundary.** Released Skill-SP/SkillRL expose representation-sensitive control surfaces; positive/negative support geometries and matched baselines isolate realizability from overlap and poor tuning. AutoSkill P19 supplies one causally dissected representation→access→behavior witness, while outcome-blind heldout qualification supports Local STRI and a preregistered behavior pilot stops without establishing propagation beyond P19.

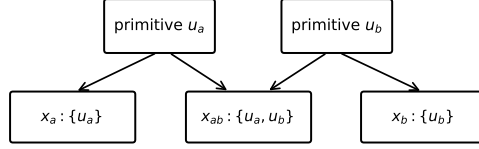
## 2 RELATION TO EXISTING WORK

**Representation, routing, and compression change different objects.** SkillsBench and Demystifying Agent Skills vary availability, presentation, representation, and retrieval difficulty (Xu & Wu, 2026; Jiang et al., 2026); SkillRouter/AgentSkillOS study large-pool selection (Zheng et al., 2026; Li et al., 2026); composition work maps task decompositions to multiple skills (Gao, 2026). SkillZip instead compresses a typed skill contract, while SkillZip Pro preserves routing and progressive-loading structure across a bundle (Bai et al., 2026a;b). These works show that representation and artifact structure matter, but do not ask whether a fixed access mechanism preserves semantic access when support/payload are held fixed and only package realization changes. External relevance/oracle-set graphs are analogues only, never executable support truth.

**Self-evolving controllers are the closest substrate.** Skill-SP (Huang et al., 2026) is closest: it samples a skill before task generation and releases independent validators. STRI then overgrants its package controller, allowing any nonnegative mass under the full frozen support matrix. SkillRL (Xia et al., 2026) supplies an identity-sensitive fixed-budget retrieval phenotype but no independent  $A$ ; AutoSkill (Yang et al., 2026) supplies one bounded behavioral seam. Work on redundancy, failures,

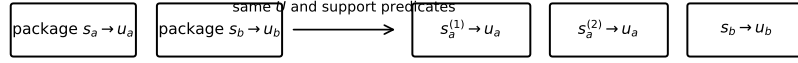
Skill-taxonomy representation is a nuisance variable, not a new semantic capability

**Frozen semantic basis (unchanged)**



**Taxonomy  $T$**

**Semantics-preserving reparameterization  $\phi(T)$**



uniform package control:  $p = (1/2, 1/2)$

same primitive support; one extra package identity

primitive mass:  $m(u_a) = 1/2, m(u_b) = 1/2$

uniform package control:  $m(u_a) = 2/3, m(u_b) = 1/3$

STRI principle: semantic control should not change solely because package representation changed

support geometry with  $R^*(A) \setminus R^* = 1$ : equalizable;  $R^* > 1$ : irreducible residual

Figure 1: STRI separates semantic capability from package representation. The conceptual identity-split example leaves the primitive basis and support predicates unchanged, yet a package-indexed controller can change primitive mass solely because one primitive has two package identities. Exact clones motivate the nuisance channel; the decisive released residual in Sec. 6 survives clone reasoning and is characterized by non-equalizable partial-overlap support geometry.

routing, and skill/harness co-evolution motivates the setting but not this support-fixed audit (Zhu et al., 2026; Liu et al., 2026; Mao et al., 2026; Dong et al., 2026; Chen et al., 2026; Wang et al., 2026).

**Optimization boundary.** Fair-exposure methods allocate attention (Singh & Joachims, 2018; Biega et al., 2018); our covering/cone/duality machinery is classical (Lovász, 1975; Sinkhorn & Knopp, 1967; Radunović & Le Boudec, 2007; Beck & Fiala, 1981; Boyd & Vandenberghe, 2004). We claim the systems invariant, information boundary, and audit mapping—not a new LP, cone theorem, or fairness objective.

### 3 PROBLEM FORMULATION: STRI AT SKILL-ACCESS BOUNDARIES

#### 3.1 RUNTIME OBJECT AND SEMANTIC INVARIANT

At access opportunity  $t$ , let  $H_t$  denote the complete pre-access agent/controller state, including task/history, prior observations and actions, already loaded skill state, and non-skill model context. Let  $U_t$  be an optional access request or control signal: an explicit query, a planned subtask, an actor-issued `load_skill` request, a router signal, or null. Under representation  $r$ , let  $P_t^{(r)}$  be the physical package realization,  $B_t$  the frozen resource budget, and  $\xi_t$  paired stochastic state. A fixed access mechanism  $\mathcal{A}_\theta$  exposes

$$E_t^{(r)} = \mathcal{A}_\theta(H_t, U_t, P_t^{(r)}, B_t, \xi_t).$$

STRI compares a preregistered semantic projection  $\phi(E_t^{(r)})$ , not raw package IDs. A semantics-preserving representation intervention keeps the underlying semantic candidate support and semantic payload fixed while allowing package identity, count, partition, multiplicity, and wrapper structure to change.

Representation-derived access state is part of the treatment path: embeddings/indexes, package metadata/router records, manifests, and representation-conditioned caches are rebuilt per arm by the same frozen construction procedure, model/configuration, and paired randomness rather than frozen across arms. **Local STRI** requires semantically equivalent package realizations to expose the same  $\phi(E_t)$  at matched  $(H_t, U_t, B_t, \xi_t)$  and fixed semantic candidate support. P19 is one prefetch instance; per-turn retrieval and progressive loading are other schedules. If Local STRI is violated,

downstream trajectories may legitimately diverge; the current paper establishes one bounded P19 downstream witness and does not establish behavioral propagation beyond P19.

### 3.2 STRUCTURAL SPECIALIZATION: PACKAGE-ONLY EXPOSURE

For the released pre-task controllers with independently defined support truth, package  $j$  contributes support column  $A_{:j}$  and package mass  $w$  induces structural semantic exposure  $Aw$ . This is an offline specialization of the general runtime object above, not a claim that every skill-using agent computes  $Aw$ . In Fig. 1, an exact clone is a duplicated support column: it may perturb identity-indexed package mass, but it should not create a new semantic degree of freedom. STRI-Cert separates two structural questions: exact identity refinements should vanish after quotienting clones; afterward, a target  $q$  is package-realizable exactly when it lies in the nonnegative cone of the remaining columns.

The five-step structural audit is: **freeze support/target provenance**  $\rightarrow$  **quotient exact clones**  $\rightarrow$  **overgrant arbitrary**  $w \geq 0$  **with full frozen**  $A \rightarrow$  **solve primal/dual** ( $R^* = 1$  constructive weighting;  $R^* > 1$  tight residual plus witness)  $\rightarrow$  **separate diagnosis from repair**. Semantic-first control can escape cone infeasibility only by changing the action basis, so it is a constructive boundary witness rather than validated downstream repair.

### 3.3 FROZEN SUPPORT MATRIX

Let  $X = \{x_1, \dots, x_n\}$  be a finite set of semantic contexts or tasks with independently defined first-party support truth, and let  $S = \{s_1, \dots, s_m\}$  be the current skill packages. Define the binary incidence matrix

$$A_{ij} = \mathbf{1}\{s_j \text{ supports } x_i\}.$$

Support is frozen before STRI: released Skill-SP validators define the tool-call matrix, while released compiler specifications plus fixed CSP/contract checks define the logical matrix. Certificates cover only rows with at least one supported package; zero-support rows are reported separately rather than misclassified as representation distortion.

At a frozen pre-task state, package-only control assigns  $w \in \mathbb{R}_{\geq 0}^m$  and induces eligibility exposure  $e = Aw$ . Exposure is control opportunity, not task probability or utility. The target  $q \in \mathbb{R}_{> 0}^n$  is a *representation-independent semantic target*: it cannot change because packages are renamed, cloned, split, or regrouped. Without an externally justified preference we use neutral  $q = \mathbf{1}$ .

### 3.4 SAME-INFORMATION PRE-TASK CONTROLLERS REDUCE TO PACKAGE MASS

In released Skill-SP, skill selection occurs before task generation. At this decision point the action is one current package identity. For any fixed pre-task state  $z$ , every randomized same-information rule—including a bandit, MLP, or neural router—induces a categorical distribution over packages, hence one nonnegative mass vector  $w(z)$ . The strongest same-information baseline is therefore arbitrary nonnegative package mass with access to the complete frozen support matrix.

This is an intentional overgrant: the baseline may ignore the released scoring rule and choose any  $w(z)$  using complete frozen  $A$  and the same pre-task information. If this class has  $R^*(A; q) > 1$ , deduplication, retuning, bandits, or neural package routers cannot remove the residual without new information or a different action basis. The reduction is pointwise in  $z$ ; it neither collapses a longitudinal controller to one static vector nor permits conditioning on the future task.

### 3.5 EXACT-REFINEMENT STRI FACTORS THROUGH A QUOTIENT

Let  $\mathcal{R}$  be a package representation and let  $Q(\mathcal{R})$  collapse exact semantic clones—packages with identical semantics, support, and controller-relevant state—into classes. Let  $G_{\mathcal{R}}$  sum a package-mass vector over these classes. For a fixed pre-context state  $z$ , define *exact-refinement STRI* by

$$G_{\mathcal{R}P\mathcal{R}}(z) = G_{\mathcal{R}'P\mathcal{R}'}(z) \quad \text{whenever } Q(\mathcal{R}) = Q(\mathcal{R}').$$

**Proposition 1 (quotient-factorization characterization).** A controller family satisfies exact-refinement STRI iff there exists a quotient-level map  $\bar{p}$  such that  $G_{\mathcal{R}P\mathcal{R}}(z) = \bar{p}(Q(\mathcal{R}), z)$  for every representation and state.

**Proof.** The forward direction defines  $\bar{p}$  using any representative of a quotient class; STRI makes this definition representative-independent. The reverse direction is immediate because equal quotients give equal  $\bar{p}$ .  $\square$

**Corollary 1 (identity-local normalization is clone-sensitive).** Suppose  $p_j = g(h_j, z) / \sum_k g(h_k, z)$  with  $g > 0$  and an exact clone receives the same local state  $h_j$ . Writing its score as  $a > 0$  and all other scores as  $B > 0$ , the semantic-class mass changes from  $a/(a+B)$  to  $2a/(2a+B)$ , an increase of  $aB/((a+B)(2a+B)) > 0$ . Repeated clones drive class mass toward one, so exact-refinement invariance requires class-aware conservation or another quotient-factorized rule.

**Proposition 2 (semantic-first construction handles arbitrary overlap).** Normalize a target ray  $q > 0$  to  $\pi = q/(1^\top q)$ . If every target-positive semantic unit  $i$  has at least one supporting package, choose any row-stochastic kernel  $K$  with  $K_{ij} = 0$  whenever  $A_{ij} = 0$ , then sample  $i \sim \pi$  before sampling implementation  $j \sim K_i$ . The semantic marginal is exactly  $\pi$  for arbitrary overlap, and any clone/split can redistribute  $K_i$  within the corresponding implementation class without changing that marginal.

**Proof.** The joint mass is  $J_{ij} = \pi_i K_{ij}$ , so  $\sum_j J_{ij} = \pi_i \sum_j K_{ij} = \pi_i$ ; support-constrained row-stochastic  $K$  exists exactly when every positive-target row is covered.  $\square$  This changes the action interface: it requires representation-independent semantic intent and conditional implementation. It is a constructive action-basis witness for escaping package-first infeasibility, not a same-information baseline or validated downstream method.

## 4 TARGET REALIZABILITY AND STRI-CERT

### 4.1 TARGET-CONDITIONED FINITE-SUPPORT CRITERION

Define the multiplicative distortion from the target ray  $q$  to the package support cone by

$$R^*(A; q) = \min_{w \geq 0, t \geq 0} t \quad \text{s.t.} \quad q \leq Aw \leq tq. \quad (1)$$

We write  $R^*(A) \equiv R^*(A; 1)$  for the neutral audit used in our released-system tables.

**Theorem 1 (target realizability in the support cone).** For any  $q > 0$ ,  $R^*(A; q) = 1$  if and only if  $q \in \text{cone}(A) = \{Aw : w \geq 0\}$ . Otherwise  $R^*(A; q) > 1$  is the tight minimum componentwise multiplicative distortion from  $q$  achievable by package-only mass.

**Proof.** The constraints force  $t \geq 1$ . At  $t = 1$  they reduce exactly to  $Aw = q$ , which is equivalent to  $q \in \text{cone}(A)$ . For any feasible  $w$ , the smallest admissible  $t$  is  $\max_i (Aw)_i / q_i$  after scaling  $w$  so that  $\min_i (Aw)_i / q_i = 1$ ; minimizing over  $w$  gives the tight distortion.  $\square$

This separates STRI from a uniformity assumption:  $q = 1$  is the neutral target, not the definition of representation invariance. It also makes exact clone/split invariance immediate: duplicating a support column does not change  $\text{cone}(A)$ , so it cannot change  $R^*(A; q)$  for any fixed  $q$ . The cone-membership statement itself is classical convex feasibility; STRI contributes the system invariant that makes this feasibility question the relevant audit object for a self-evolving skill controller, not a new theorem about convex cones.

**Proposition 3 (dual certificate).** The dual of Eq. (1) is

$$D^*(A; q) = \max_{\alpha, \beta \geq 0} q^\top \alpha \quad \text{s.t.} \quad A^\top (\beta - \alpha) \geq 0, \quad q^\top \beta \leq 1. \quad (2)$$

Strong duality gives  $D^*(A; q) = R^*(A; q)$ . Thus every dual-feasible pair is a certified lower bound, while an optimum is complete even when no local witness template is available. On every audited neutral-target regime the numerical primal–dual gap is below  $10^{-8}$ .

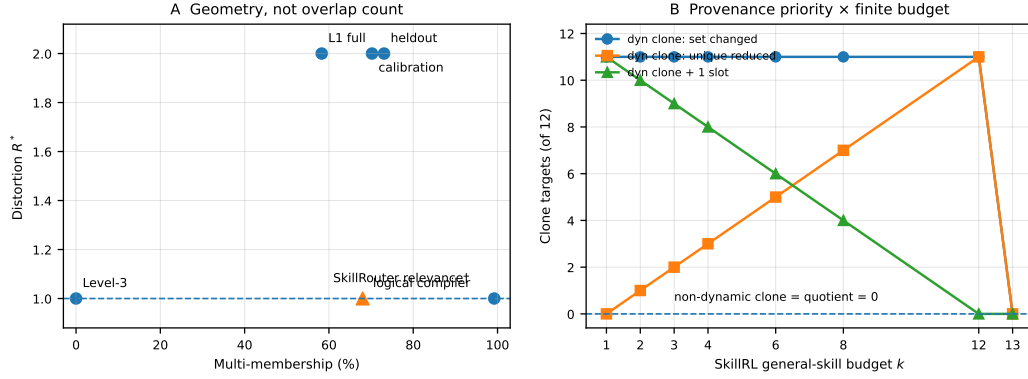


Figure 2: Boundary breadth. **A:** support geometry separates residual/equalizable regimes; SkillRouter<sup>†</sup> is an expert relevance analogue, not executable support. **B:** SkillIRL exact clones change semantic retrieval only under fresh-identity priority plus finite budget; placebo/quotient controls stay at zero and capacity closes the semantic-set effect.

#### 4.2 HUMAN-READABLE FACTOR-2 WITNESS

**Theorem 2 (global-singleton overlap lower bound).** Suppose packages  $a, b$  each have a context supported by exactly that one package over the full frozen support universe, and another context is supported by both  $a$  and  $b$ , possibly with additional packages. If both singleton contexts retain positive exposure, then every nonnegative package mass has max/min exposure ratio at least two on the focal rows, and therefore  $R^*(A) \geq 2$  for the full support matrix.

**Proof.** Let  $r > 0$  be the minimum exposure among the two singleton contexts and the shared context. Since the first singleton support signature is exactly  $\{a\}$ , its exposure is  $w_a \geq r$ ; likewise  $w_b \geq r$ . The shared context contains both  $a$  and  $b$ , and additional packages have nonnegative weight, so its exposure is at least  $w_a + w_b \geq 2r$ . Thus every  $w$  has focal max/min ratio at least two. For the same  $w$ , the full-matrix maximum is no smaller than the focal maximum and the full-matrix minimum is no larger than the focal minimum, so the full-matrix ratio is also at least two. Minimizing over  $w$  gives  $R^*(A) \geq 2$ .  $\square$

For the neutral target  $q = 1$ , Theorem 2 is also a sparse dual certificate: put  $\alpha = 1$  on the two singleton rows,  $\beta = 1$  on their shared row, and zero elsewhere. Dual feasibility follows from the support pattern and the objective is two. In released Skill-SP Level-1, the LP dual recovers such a sparse optimum and the primal attains  $R^* = 2$ .

Two released Skill-SP instances realize this sparse factor-2 witness; the supplementary visualization reports both support patterns.

#### 4.3 STRI-CERT AND DEPLOYABLE VARIANTS

STRI-Cert is a diagnostic protocol, not a learned predictor or a new LP solver. Its output has three operational meanings: **realizable** ( $R^* = 1$ ) means package-only retuning is sufficient *in principle* to hit the declared target; **residual** ( $R^* > 1$ ) means no same-information package router can hit that target without changing information, assumptions, or the action basis; and **fail-closed** means support/target provenance or optimization is not valid enough to make either statement. The dual may expose a sparse obstruction, while the primal gives a constructive weighting in the realizable case. HiGHS solves both programs with paper-level tolerance  $10^{-8}$ ; audited primal–dual gaps are at most  $10^{-8}$ . The certificate concerns feasibility, not downstream utility or a unique weight vector.

Two variants make deployment assumptions explicit. With independently justified graded support  $0 \leq L \leq A \leq U$ , the box-robust audit uses  $Lw \geq q$  and  $Uw \leq tq$ ; estimating valid bounds remains an identification problem. Adding  $w_j \leq \rho \sum_k w_k$  audits package-mass concentration via  $R_\rho^*(A; q) \geq R^*(A; q)$ . Both remain linear programs.

Table 1: Two-stage audit across released systems. We test observable identity sensitivity first and run STRI-Cert only when independent support truth exists; absence of a support contract is not repaired with candidate-defined labels.

| System   | Identity perturbation                                                                                                                                         | Independent $A$ ?            | Audit role                                                       |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------|
| Skill-SP | Pristine sampler gives $2\times$ eligibility; sampler-input clone changes class mass $1/15 \rightarrow 1/8$ , while quotient conservation restores it exactly | Yes: released validators     | Controller violation + exact clone repair + geometry certificate |
| SkillIRL | Fresh-ID exact clones change semantic retrieval for 11/12 targets across $k = 1-12$ ; effect vanishes at $k = 13$                                             | No explicit support contract | Budget-bound phenotype + placebo/quotient controls; no $R^*$     |

Table 2: Practical package-weight baselines. Entries are neutral-target worst-case exposure ratios (lower is better). “Heldout” is the frozen 4-tool $\rightarrow$ 4-tool split; “LOO max” is the worst of eight leave-one-tool-out transfers with no heldout refit. Semantic-first changes the action basis and is only an upper bound.

| Baseline                        | L1 full     | Heldout     | LOO max     | L3          | Logical     |
|---------------------------------|-------------|-------------|-------------|-------------|-------------|
| Released uniform                | 2.00        | 2.00        | 2.00        | 1.00        | 8.00        |
| Inverse support size            | 98.33       | 2.89        | 3.50        | 3.00        | 15.67       |
| Inverse $\sqrt{\text{support}}$ | 10.87       | 2.38        | 2.58        | 1.73        | 11.02       |
| NNLS target fit                 | 5.51        | 3.98        | 6.10        | 1.00        | 1.00        |
| Exact min-cover + uniform       | 2.00        | 2.00        | 2.00        | 1.00        | 1.00        |
| Max-min fair                    | 2.00        | 2.00        | 2.00        | 1.00        | 1.00        |
| Exact $R^*$                     | <b>2.00</b> | <b>2.00</b> | <b>2.00</b> | <b>1.00</b> | <b>1.00</b> |
| Semantic-first <sup>†</sup>     | 1.00        | 1.00        | 1.00        | 1.00        | 1.00        |

## 5 EXPERIMENTAL SETUP

**RQ1: does equivalent packaging change semantic access?** We first localize representation sensitivity before asking about utility. Skill-SP supplies a pre-task package-control seam, SkillIRL supplies finite-budget identity-sensitive retrieval, and AutoSkill supplies a first-party search boundary with exact split/placebo/quotient controls. The primary readout is semantic access or exposure under matched underlying capability, not raw package IDs and not task success.

**RQ2: is the residual structural rather than a duplicate/overlap/tuning artifact?** On frozen support matrices we compute neutral  $R^*(A)$  with primal/dual tolerance  $10^{-8}$ , compare practical package-weight baselines, transfer calibration-fitted weights without heldout refitting, and include both residual and equalizable support geometries. Level-1 is the main positive regime; Level-3 and the logical compiler provide natural negative boundaries. Target rays, edit radii, witness peeling, degree-preserving rewiring, and solver timing are supporting robustness analyses on frozen artifacts, not additional independent claims.

**RQ3: does local access divergence propagate downstream?** AutoSkill P19 is the only current behavior-level positive witness. We pair its split/placebo/quotient result with a specific-mediator add-back versus matched cleanup control. Separately, an outcome-blind heldout qualification identifies representation-sensitive retrieval units before behavior; the preregistered behavior pilot stops when its split-specific gate is not met. This STOP is a claim boundary rather than evidence of a population no-effect.

**Evidence discipline.** The structural  $R^*$  path and runtime access path share a semantic-space motivation but are not one causal mechanism. External SkillRouter/AgentSkillOS graphs are labelled analogues rather than executable support truth; a separate SkillsBench qualification rejects task-local availability as exact support (79/87 rows disagree with `required_skills`). New breadth analyses use frozen public artifacts only—zero new annotations, model calls, or GPU runs.

Table 3: Neutral-target boundary across first-party support regimes. High multi-membership is neither necessary nor sufficient for a structural residual.

| Regime                        | Covered | Multi | Overlap | $R^*$ | Certificate |
|-------------------------------|---------|-------|---------|-------|-------------|
| API-Bank Level-1 full         | 314     | 183   | 58.3%   | 2.0   | Residual    |
| Level-1 calibration tools     | 47      | 33    | 70.2%   | 2.0   | Residual    |
| Level-1 tool-disjoint heldout | 52      | 38    | 73.1%   | 2.0   | Residual    |
| API-Bank Level-3              | 34      | 0     | 0.0%    | 1.0   | Equalizable |
| Logical compiler validation   | 128     | 127   | 99.2%   | 1.0   | Equalizable |

## 6 EXPERIMENTS AND RESULTS

The results follow the three questions above: first local/control sensitivity, then the structural boundary, then downstream propagation. Robustness and solver-cost analyses are reported only after the load-bearing RQ results.

### 6.1 RQ1–RQ2: SKILL-SP CONTROL SENSITIVITY AND A RESIDUAL SUPPORT GEOMETRY

API-Bank Level-1 has 314 covered rows, 183 with multiple memberships, and neutral  $R^* = 2$ . The released pre-task sampler initializes all 15 tool-call packages at probability  $1/15$ , so a double-supported row receives  $2/15$  eligibility versus  $1/15$  for a singleton. The source paper reports a  $-1.9$  point accuracy change when dynamic routing is replaced by uniform routing (Huang et al., 2026); we use this only to establish that allocation is operational, not to attribute utility to STRI.

For a same-state sampler-input reparameterization, we copy one active package with identical content, support, statistics, and sampler state under a fresh ID and rerun the author’s weighting function. This is not a claim that the induction path would admit a literal duplicate. The cloned semantic class moves from  $1/15$  to  $2/16 = 1/8$ , yielding prompt-mixture  $TV = 7/120 = 0.0583$  because package ID is absent from the questioner message. Quotient-conserved mass instead splits the original  $1/15$  class mass and restores prompt TV exactly to zero. These are input-mixture statements, not generated-task or utility claims.

A frozen tool-disjoint test gives the same structural answer: calibration has 47 covered rows/33 multi-memberships and  $R^* = 2$ ; heldout has 52/38 and  $R^* = 2$ . In the 4-tool→4-tool split, calibration-fitted uniform, exact min-cover, max–min, and exact- $R^*$  weights retain heldout ratio 2, while inverse-support and NNLS transfer at 2.89 and 3.98. A stricter eight-fold leave-one-tool-out audit refits on seven tools and freezes weights for the eighth: uniform/exact-min-cover/exact- $R^*$  remain 2.0 on every fold, whereas NNLS has median 2.62 and worst 6.10. Thus the residual is not a single-split refitting artifact.

### 6.2 RQ2: LEVEL-3 DISJOINT-SUPPORT NEGATIVE CONTROL

Level-3 provides a natural released negative regime. Thirty-four rows are covered, every covered row has exactly one skill membership, and  $R^* = 1$ . Equal weights on the active packages equalize every covered task. STRI-Cert therefore does not mechanically label every released taxonomy problematic.

### 6.3 RQ2: LOGICAL COMPILER—HIGH OVERLAP BUT NO RESIDUAL

The first-party logical release gives a stronger negative control: 128/128 programmatically generated units pass the author’s compiler/uniqueness contracts and are cross-checked against all eight released support specifications, with no language model. Although 127/128 units have multiple memberships across 38 support patterns,  $R^* = 1$  because `skill_008` supports every unit. This equalizer is concentrated: the max-share audit gives  $R_{0.75}^* = 4/3$ ,  $R_{0.5}^* = 2$ , and  $R_{0.25}^* = 4$ . High overlap is therefore not sufficient; realizability depends on support geometry and the declared feasible package-mass set.

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#### 6.4 RQ1 BOUNDARY: SKILLRL FINITE-BUDGET IDENTITY SENSITIVITY

SkillRL supplies a complementary identity-sensitive retrieval seam. Across 223 ALFWorld descriptions and all 12 exact-clone targets, fresh-dyn\_ priority changes semantic retrieval for 11/12 targets for every  $k \in \{1, 2, 3, 4, 6, 8, 12\}$ ; unique-content crowd-out grows from 0 to 11 targets. A same-content non-dynamic-ID duplicate and exact-semantic quotient cause zero semantic-set changes at every tested budget. At  $k = 13$ , where all 12 static contents plus the clone fit, the semantic-set effect vanishes; one extra slot at  $k = 12$  also restores it. This is a provenance-priority $\times$ finite-budget phenotype, not an  $R^*$  or utility claim; the separate qualified endpoint bridge is not a population-level no-effect theorem. External SkillRouter/AgentSkillOS relevance graphs are retained only as non-executable analogues in the supplement and related-work boundary.

#### 6.5 RQ1–RQ3: AUTOSKILL LOCAL ACCESS AND BOUNDED BEHAVIORAL PROPAGATION

AutoSkill (Yang et al., 2026) exposes a first-party `sdk.search` seam. In one frozen five-skill P19 library at `top_k=5`, a four-way exact-semantic split crowds out a post-checkout skill; ID renaming preserves, and exact-semantic quotienting restores, the original semantic set. The mechanical post-checkout behavior occurs in 6/6 original, 0/6 split, 3/3 placebo, and 3/3 quotient runs (one-sided Fisher  $p = 0.00108$ ). Under the split, adding the crowded-out skill back restores behavior in 3/3 fresh runs versus 0/3 for a matched cleanup add-back ( $p = 1/20$ ). This isolates one mediator on one substrate.

An outcome-blind preflight qualified 9/9 held-out units; SHA-256 selected two cross-scenario/position units before behavior. Across 8/8 valid A/B/C/D runs, neither met the frozen five-dimensional split-specific gate ( $A=B=C=D$  in one;  $A=B=C \neq D$  in the other), so repeat-2 and the remaining seven units were stopped. The pilot therefore does not establish behavioral propagation beyond P19; P19 remains the single bounded downstream witness.

#### 6.6 RQ2 SUPPORTING ANALYSES: MATCHED BASELINES, ROBUSTNESS, AND COST

On frozen support, Level-1 uniform attains optimum 2; inverse-support reaches 98.33 (10.87 tempered), while NNLS lowers CV 0.312 $\rightarrow$ 0.215 but worsens ratio to 5.51. Exact enumeration shows budgets 1–2 cannot cover Level-1; `skill_003/004/015` attain unrestricted  $R^* = 2$  at budget 3, unchanged through 6. Level-3 needs four packages for  $R^* = 1$ ; universal `skill_008` alone gives logical  $R^* = 1$ . Thus average fit and controller sparsity do not explain the residual.

**Supporting robustness and cost.** Supplementary artifacts retain the broader frozen checks: Level-1 stays residual under 1,387 additions, 366 valid deletions, and 200 degree-preserving bipartite rewirings; exact neutral-target MILPs require at least 22 added cells or 71 deletions; witness peeling finds 22 pairwise-disjoint three-row witnesses spanning 66 rows/19 tools (21.0% of Level-1) before  $R^* = 1$ . Dense HiGHS takes 0.00234 s at  $128 \times 8$  and 0.685 s at  $16,384 \times 96$ . These analyses support robustness/tractability only and do not create additional main claims.

### 7 DISCUSSION, LIMITATIONS, AND TAKEAWAY

The main result is layered rather than monolithic. At the runtime layer, package representation can change semantic access even when the underlying capability is unchanged; outcome-blind heldout qualification shows that this local sensitivity is not unique to P19. At the structural layer, support geometry—not overlap prevalence or average fit—separates equalizable and residual regimes:  $R^* = 1$  permits package-only retuning in principle, whereas  $R^* > 1$  excludes that control class for the declared target. These layers are complementary but neither predicts the other.

Downstream propagation is deliberately narrower. AutoSkill P19 provides one causally dissected representation $\rightarrow$ access $\rightarrow$ mediator $\rightarrow$ behavior witness, while the preregistered heldout behavior pilot stops without establishing propagation beyond P19. The AutoSkill path and SkillRL phenotype therefore support neither population utility, safety, nor regret claims, and semantic-first control remains a constructive action-basis witness rather than a validated repair. A non-clone repeated-access successor under representation-neutral capacity is future claim expansion, not a requirement for this narrow result.

## AI USE STATEMENT

In this work, generative AI tools assisted with conceptual framing and hypothesis refinement; formulation and critique of mathematical claims and proof drafts; experiment and falsifier design; implementation and review of analysis, experiment-control, and manuscript-quality-assurance code; interpretation of intermediate results; primary-source literature triage; and preparation or editing of manuscript text, figures, tables, and references. Generative models were also used as the task proposer in a preregistered Qwen3-8B dynamic pilot; that pilot failed its frozen author-native qualification gate, remains documented in the supplement, and is excluded from scientific claims about dynamic STRI. Load-bearing numerical claims in the submitted narrow paper are bound to released first-party artifacts or deterministic/programmatic analysis outputs. The support certificate, theorem edge cases, transition logic, and paper-level numerical/citation constraints were exercised by executable tests; related-work boundaries and bibliographic metadata were checked against primary or official sources. AI-assisted outputs were not treated as independent scientific authority. We take responsibility for the final content of this work, including text, claims, code, and artifacts produced with the aid of generative AI.

## REPRODUCIBILITY STATEMENT

The main paper states the skill-access-boundary runtime object, the rule that representation-derived access state is rebuilt per arm, the exact-refinement quotient characterization, semantic-first design construction, target-conditioned primal/dual programs, closed-form witness, box-robust extension, and constrained-controller audit used for the load-bearing claims. The Skill-SP controller audit is bound to author commit `bb693c89fee66e1f824d6a777759a49b7a295a83`, released `library.py` SHA-256 `8441de35...e2756e0`, question-generation SHA-256 `1b4cb67f...82d739b`, initial package-bundle SHA-256 `d4c4350a...68b6e7`, and tool-call membership SHA-256 `b73d8637...6f2e17d4`. The supplement packages our controller-audit/certificate code, tests, frozen receipts and matrices, but intentionally does not redistribute the third-party Skill-SP repository; its README gives the exact author commit required for end-to-end rerun. LPs use SciPy HiGHS; certificate classification uses tolerance  $10^{-8}$ . The offline completion analysis is implemented in `asset_first_stri_offline_completion_analysis_20260824.py` and records the frozen Level-1 structured-misspecification families plus deterministic synthetic  $R^*$  CPU profiling in a content-addressed JSON artifact. Executable tests cover the target-conditioned dual, target-ray scaling, semantic-first target realization under arbitrary overlap, quotient clone invariance, sampler-weight recomputation, literal-duplicate rejection, exact questioner-message equality, the 7/120 prompt-mixture shift, quotient restoration to zero, one-cell support perturbations, per-tool LP results, and the offline analysis contract. The added E4/E6 analyses use no new support annotations, task outcomes, model calls, or GPU runs. All dynamic thresholds were frozen before outcomes. The failed Qwen3-8B bank remains a transparent non-result; the qualified SkillRL bridge separately records 18/24 competence calibration, 24 paired A/B/C/D units, exact A/D coupling, and the statistical-resolution boundary that prevents upgrading a realization STOP into a population-level no-effect claim.

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